

Using PyTorch for Natural Language Processing

This article is focused on the classes that PyTorch provides, which help with Natural Language Processing (NLP).

Natural language processing (NLP) is a subset of computer science, and is mainly about artificial intelligence (AI). It enables computers to understand and process human

language. Technically, the main objective of NLP is to program computers for analysing and processing natural language data.

PyTorch is one of the most popular deep learning frameworks that is

based on Python and is supported by Facebook. It is an open source machine learning library built on the Torch library, and used for applications such as computer vision and natural language processing. Developed by Facebook's AI Research (FAIR) lab, it is free and open source software released under the modified BSD licence. Although the Python interface is more polished and the main focus of development, PyTorch also has a C++ interface. Some examples of PyTorch are Uber's Pyro, HuggingFace's Transformers, etc.

PyTorch provides two high-level characters:

- Tensor computing (like NumPy) with strong acceleration through graphics processing units (GPU)
- Deep neural networks built on a tape based automatic differentiation (autodiff) system

How to use PyTorch for NLP

There are mainly six classes in PyTorch that can be used for NLP related jobs using recurrent layers:

- `torch.nn.RNN`
- `torch.nn.LSTM`
- `torch.nn.GRU`
- `torch.nn.RNNCell`
- `torch.nn.LSTMCell`
- `torch.nn.GRUCell`

Understanding these classes, their parameters, their inputs and their outputs is key to getting started with building your own neural networks for NLP in Pytorch.

Unrolled diagrams are used to provide simple-to-grasp explanations

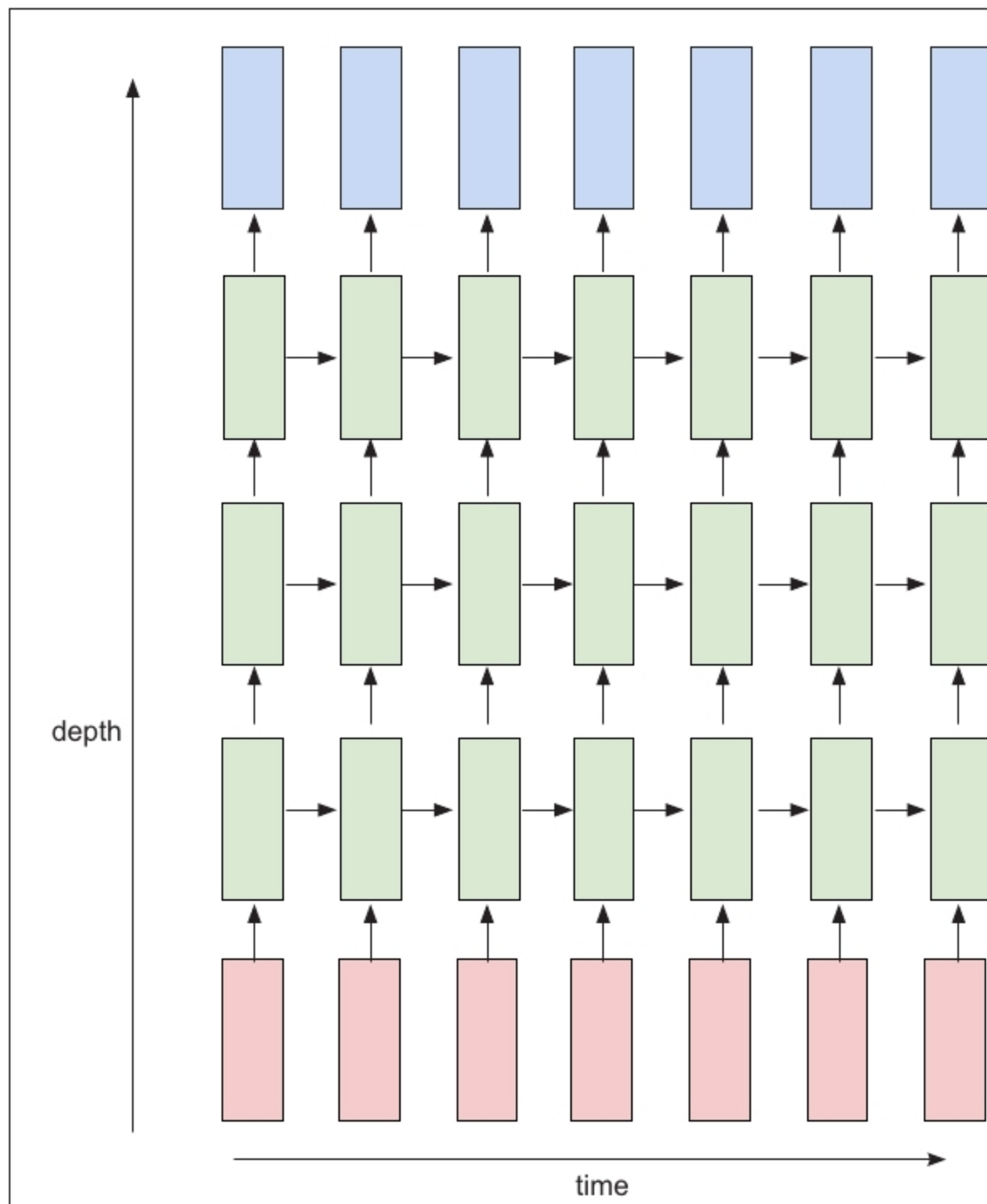


Figure 1: Recurrent neural network

